

1) Trigonometry

a) Exact values

The unit circle can be used to define the trigonometric ratios.

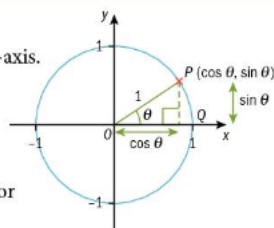
The angle θ is measured in an anticlockwise direction from the positive x -axis.

$\cos \theta$ can be defined as the x coordinate of P .

$\sin \theta$ can be defined as the y coordinate of P .

$\tan \theta$ can be defined as $\frac{\sin \theta}{\cos \theta}$.

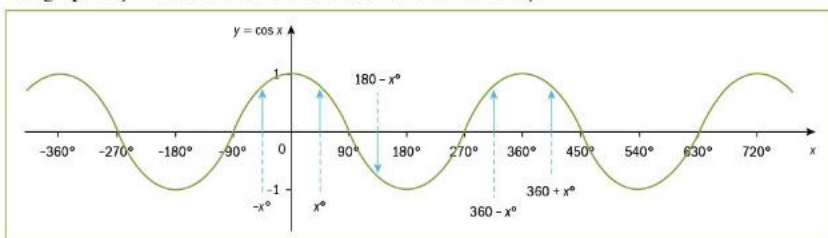
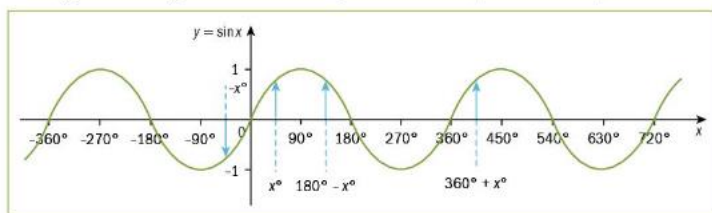
The diagram can be used to write down the values of $\sin$, $\cos$ and $\tan$ for 0° , 90° , 180° , 270° and 360° .



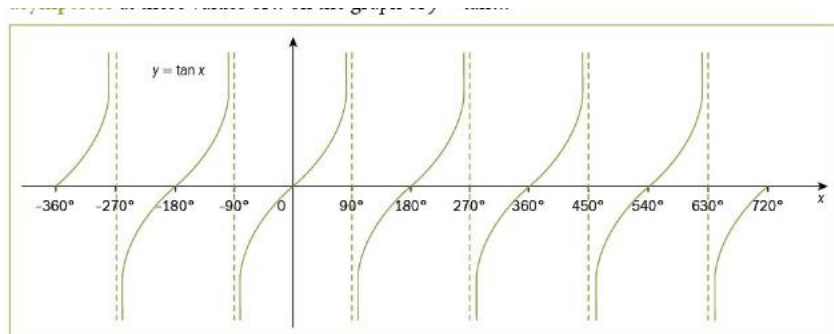
Note:

b) Graph of trigonometric functions

We say $y = \sin x$ is periodic and has a **period** of 360° (or 2π radians).



$$\cos(-x) = \cos x$$



Example 3

Find all the angles x (to the nearest degree), where $0^\circ < x < 360^\circ$, such that $\sin x = 0.88$.

$$x = \sin^{-1}(0.88) \\ = 62^\circ$$

Use a calculator to find the principal value.

$$x = 62^\circ, 180^\circ - 62^\circ$$

Use a graph sketch to find the second angle in the stated interval.

$$x = 62^\circ, 118^\circ$$

Example 4

Solve $\tan x = 2.35$ for $0^\circ < x < 720^\circ$.

Give your answers correct to one decimal place.

$$x = \tan^{-1}(2.35) \\ = 66.9^\circ \text{ (1 d.p.)}$$

This is the principal value given by a calculator.

$$x = 66.9^\circ, 180^\circ + 66.9^\circ, 360^\circ + 66.9^\circ, 540^\circ + 66.9^\circ$$

$$x = 66.9^\circ, 246.9^\circ, 426.9^\circ, 606.9^\circ \text{ (1 d.p.)}$$

Add on multiples of 180° to give all values in the interval $0^\circ < x < 720^\circ$.

1. Find the period of each of these functions.

a) $y = \sin 2x$

b) $y = \cos 5x$

c) $y = \cos x + 1$

d) $y = 5 \sin(x - 30)^\circ$

e) $y = \tan \frac{1}{2}x$

f) $y = 2 \tan 3x - 4$

g) $y = \tan(x + 45)^\circ$

h) $y = 3 \sin(2x - 60)^\circ$

2. Sketch, on separate diagrams, the graphs of

a) $y = \sin 3x$ for $0^\circ \leq x \leq 360^\circ$

b) $y = -\cos 2x$ for $0^\circ \leq x \leq 360^\circ$

c) $y = \tan \frac{1}{2}x$ for $-360^\circ \leq x \leq 360^\circ$

d) $y = 2 \sin 4x$ for $0^\circ \leq x \leq 180^\circ$

e) $y = 3 \tan(x + 30^\circ)$ for $0^\circ \leq x \leq 360^\circ$

f) $y = 3 \sin 2x - 1$ for $0^\circ \leq x \leq 360^\circ$.

c) Trigonometric equations

Example 8

Solve the equation $4 \sin x + 3 = 5$ for $0^\circ \leq x \leq 360^\circ$.

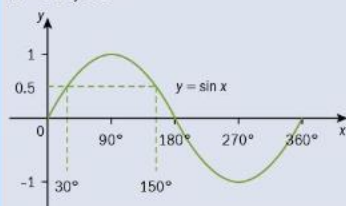
$$4 \sin x + 3 = 5$$

$$4 \sin x = 2$$

$$\sin x = \frac{1}{2}$$

$$x = 30^\circ, 180^\circ - 30^\circ$$

$$x = 30^\circ, 150^\circ$$



First find the value of $\sin x$.

Use a sketch graph to help you find all the angles in the required interval.

Example 9

Solve the equation $3 \cos 2\theta = -1$ for $0^\circ \leq \theta \leq 360^\circ$.

Values for $0^\circ \leq \theta \leq 360^\circ$ are required, so values for $0^\circ \leq 2\theta \leq 720^\circ$ will be needed.

$$3 \cos 2\theta = -1$$

$$\cos 2\theta = -\frac{1}{3}$$

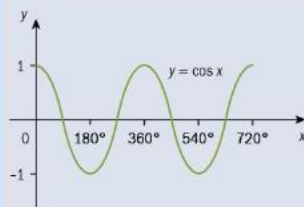
$$2\theta = 109.47\dots^\circ, \quad 360^\circ - 109.47\dots^\circ$$

$$360^\circ + 109.47\dots^\circ, \quad 720^\circ - 109.47\dots^\circ$$

$$2\theta = 109.47\dots^\circ, \quad 250.52\dots^\circ$$

$$469.47\dots^\circ, \quad 610.52\dots^\circ$$

$$\theta = 54.7^\circ, 125.3^\circ, 234.7^\circ, 305.3^\circ$$



Use your calculator to find the principal value $109.47\dots^\circ$.

Use the graph to find the other angles.

Note: You should normally give angles in degrees correct to 1 decimal place, so make sure you work with more than this to avoid rounding errors. Remember to work out all the values from the graph before you divide by 2.

1. Solve these equations for $0^\circ \leq x \leq 360^\circ$.

a) $4 \sin x = 3$	b) $5 \tan x - 1 = 9$	c) $6 \cos x + 5 = 8$
d) $8 \sin(x + 20^\circ) = 5$	e) $10 - 3 \cos x = 9$	f) $9 \sin(x - 15^\circ) = -4$
2. In the following equations, find all the values of θ between 0° and 360° .

a) $\cos \theta = 0.776$	b) $\sin 2\theta = -0.364$	c) $\tan 3\theta = 1.988$
d) $\cos \frac{\theta}{2} = -0.379$	e) $\tan \frac{\theta}{2} = -1.030$	f) $\sin \frac{3\theta}{2} = 0.664$
3. Solve the following equations for $0 \leq x \leq 2\pi$.

a) $3 \tan x + 2 = 5$	b) $\sin\left(x - \frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$	c) $\sin 2x = \frac{1}{2}$
d) $6 \cos 3x - 3 = 0$	e) $7 - 2 \tan 4x = 13$	f) $10 \cos \frac{x}{3} = 1$
4. Solve $6 \cos 30x - 3 = 0$ for $0^\circ \leq x < 24^\circ$.
5. Solve $3 \tan x = \sqrt{27}$ for $-180^\circ \leq x < 180^\circ$.
6. Solve $5 + 2 \cos\left(3\theta - \frac{\pi}{4}\right) = 6$ for $-\pi \leq \theta < \pi$.

d) Trigonometric identities

Example 10Simplify a) $\frac{\sin^3 \theta + \sin \theta \cos^2 \theta}{\cos \theta}$ b) $\frac{1 + \cos \theta}{\sin \theta} + \frac{\sin \theta}{1 + \cos \theta}$.

a) $\frac{\sin^3 \theta + \sin \theta \cos^2 \theta}{\cos \theta} = \frac{\sin \theta (\sin^2 \theta + \cos^2 \theta)}{\cos \theta}$

Factorise the numerator.

$$= \frac{\sin \theta}{\cos \theta}$$

Use $\sin^2 \theta + \cos^2 \theta = 1$.

$$= \tan \theta$$

Use $\frac{\sin \theta}{\cos \theta} = \tan \theta$.

b) $\frac{1 + \cos \theta}{\sin \theta} + \frac{\sin \theta}{1 + \cos \theta} = \frac{(1 + \cos \theta)^2 + \sin^2 \theta}{\sin \theta (1 + \cos \theta)}$

Add the fractions.

$$= \frac{1 + 2 \cos \theta + \cos^2 \theta + \sin^2 \theta}{\sin \theta (1 + \cos \theta)}$$

Multiply out $(1 + \cos \theta)^2$.

$$= \frac{1 + 2 \cos \theta + 1}{\sin \theta (1 + \cos \theta)}$$

 $\sin^2 \theta + \cos^2 \theta = 1$

$$= \frac{2(1 + \cos \theta)}{\sin \theta (1 + \cos \theta)}$$

Factorise the numerator.

$$= \frac{2}{\sin \theta}$$

Divide the numerator and denominator by $(1 + \cos \theta)$, for $\cos \theta \neq -1$.**Example 11**Prove the identity $\frac{\sin \theta}{\tan \theta (1 - \cos \theta)} = \frac{\cos \theta}{1 - \cos \theta}$, $\sin \theta \neq 0$.

$$\frac{\sin \theta}{\tan \theta (1 - \cos \theta)} = \frac{\sin \theta}{\frac{\sin \theta}{\cos \theta} (1 - \cos \theta)}$$

Start with the left hand side and replace $\tan \theta$ with $\frac{\sin \theta}{\cos \theta}$.

$$= \frac{\cancel{\sin \theta}^1 \cos \theta}{\cancel{\sin \theta}_1 (1 - \cos \theta)}$$

 $\div \frac{\sin \theta}{\cos \theta}$ is the same as $\times \frac{\cos \theta}{\sin \theta}$.

$$= \frac{\cos \theta}{1 - \cos \theta}$$

Divide numerator and denominator by the common factor $\sin \theta$.

3. Simplify

a) $\frac{\sin^3 \theta + \sin \theta \cos^2 \theta}{\sin \theta}$ b) $1 + \sin \theta + \frac{\cos^2 \theta}{1 + \sin \theta}$.

4. Prove the identity $\frac{\sin^4 \theta}{\tan \theta} = \sin \theta \cos \theta - \sin \theta \cos^3 \theta$.

5. Prove the following identities

a) $\cos^3 \theta = \cos \theta - 2 \sin^2 \theta \cos \theta + \cos \theta \sin^4 \theta$

b) $(4 \sin \theta + 3 \cos \theta)^2 + (3 \sin \theta - 4 \cos \theta)^2 = 25$

c) $\tan^2 A - \tan^2 B = \frac{\sin^2 A - \sin^2 B}{\cos^2 A \cos^2 B}$.

6. **Show that** $(1 + \sin \theta + \cos \theta)^2 = 2(1 + \sin \theta)(1 + \cos \theta)$.

7. Simplify

a) $\frac{\sin^2 \theta - \cos^2 \theta}{\sin \theta - \cos \theta}$ b) $\frac{\sin^4 \theta - \cos^4 \theta}{\sin^2 \theta - \cos^2 \theta}$ c) $\frac{\sin^4 \theta - \cos^4 \theta}{\sin \theta - \cos \theta}$.

8. Prove the following identities.

a) $\tan \theta = \sqrt{\frac{\sin^2 \theta}{1 - \sin^2 \theta}}$ b) $\frac{1 - \sin^2 \theta}{\tan \theta} = \frac{\cos^3 \theta}{\sin \theta}$.

c) $\frac{\sin^2 \theta}{1 - \cos \theta} = 1 + \cos \theta$ d) $(2 - \cos^2 \theta)^2 - 4 \sin^2 \theta = \cos^4 \theta$

9. a) Express $2 \cos x - \sin^2 x$ in the form $(\cos x + p)^2 + q$, stating the values of p and q .

b) Use your answer to part (a) to find the maximum and minimum values of $2 \cos x - \sin^2 x$.

Solve the equations in questions 1 to 8 for $0^\circ \leq \theta \leq 360^\circ$.

1. $\cos^2 \theta - \sin^2 \theta = 0$.

2. $2 \cos^2 \theta + \sin \theta = 1$.

3. $3 \sin^2 \theta = \cos^2 \theta$.

4. $\cos^2 \theta - 4 \sin^2 \theta = 1$.

5. $1 + \sin \theta \cos^2 \theta = \sin \theta$

6. $6 \sin(\theta + 70)^\circ - 5 \cos(\theta + 70)^\circ = 0$

7. $3 \cos 2\theta = 4 \sin 2\theta$.

8. $2 \sin \theta = 3 \tan \theta$.

9. Solve the equation $\sin \theta + 5 \cos \theta = 3 \sin \theta$ for $0 \leq \theta \leq 2\pi$.

10. Solve the equation $2 \cos \theta - 5 \sin \theta = 4 \cos \theta + 3 \sin \theta$ for $0 \leq \theta \leq 2\pi$.

11. a) Simplify $\frac{6 - 6 \cos^2 \theta}{2 \sin \theta}$.

b) Hence solve the equation $6 - 6 \cos^2 \theta = 3 \sin \theta$ for $0 \leq \theta \leq 2\pi$.

12. Solve the equation $\tan^3 \theta - \tan^2 \theta - 2 \tan \theta = 0$ for $0^\circ \leq \theta \leq 180^\circ$.

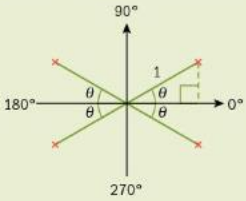
Summary:

Chapter Summary

Exact values of trigonometric functions

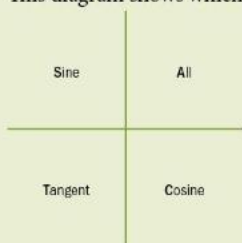
Degrees	Radians	sin	cos	tan
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$

Angles in the range 0° to 360°

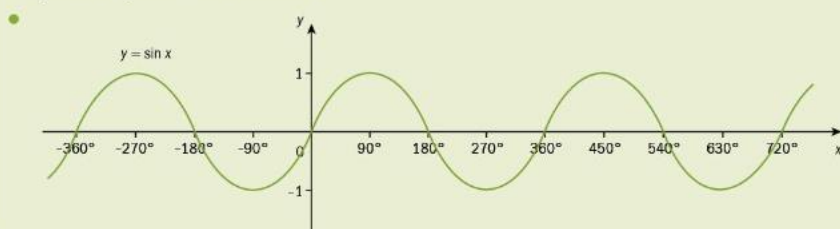


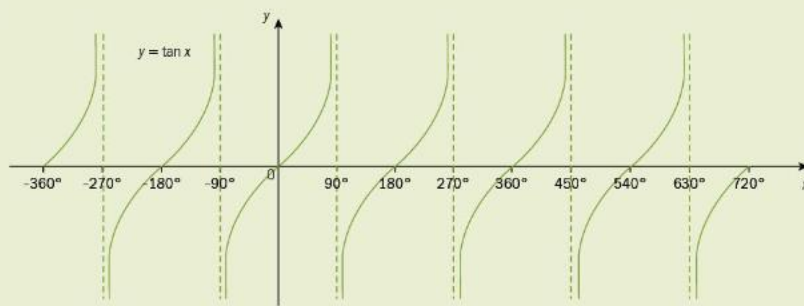
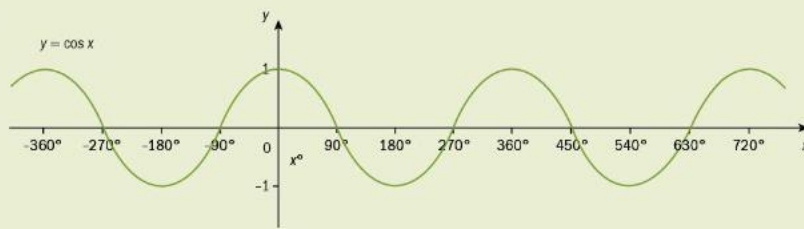
$\sin(180^\circ - \theta) = \sin \theta$
 $\cos(180^\circ - \theta) = -\cos \theta$
 $\tan(180^\circ - \theta) = -\tan \theta$
 $\sin(180^\circ + \theta) = -\sin \theta$
 $\cos(180^\circ + \theta) = -\cos \theta$
 $\tan(180^\circ + \theta) = \tan \theta$
 $\sin(360^\circ - \theta) = -\sin \theta$
 $\cos(360^\circ - \theta) = \cos \theta$
 $\tan(360^\circ - \theta) = -\tan \theta$

- This diagram shows which functions are positive in each quadrant.



Graphs of trigonometric functions





Composite graphs

- The graph of $y = \sin(x + c^\circ)$ is a translation of $y = \sin x$ by $\begin{pmatrix} -c \\ 0 \end{pmatrix}$.
- The graph of $y = \sin x + d$ is a translation of $y = \sin x$ by $\begin{pmatrix} 0 \\ d \end{pmatrix}$.
- The graph of $y = a \sin x$ is a stretch of $y = \sin x$, factor a , parallel to the y -axis.
- The graph of $y = \sin bx$ is a stretch of $y = \sin x$, factor $\frac{1}{b}$, parallel to the x -axis.

Trigonometric identities

- $\tan \theta = \frac{\sin \theta}{\cos \theta}$
- $\sin^2 \theta + \cos^2 \theta = 1$